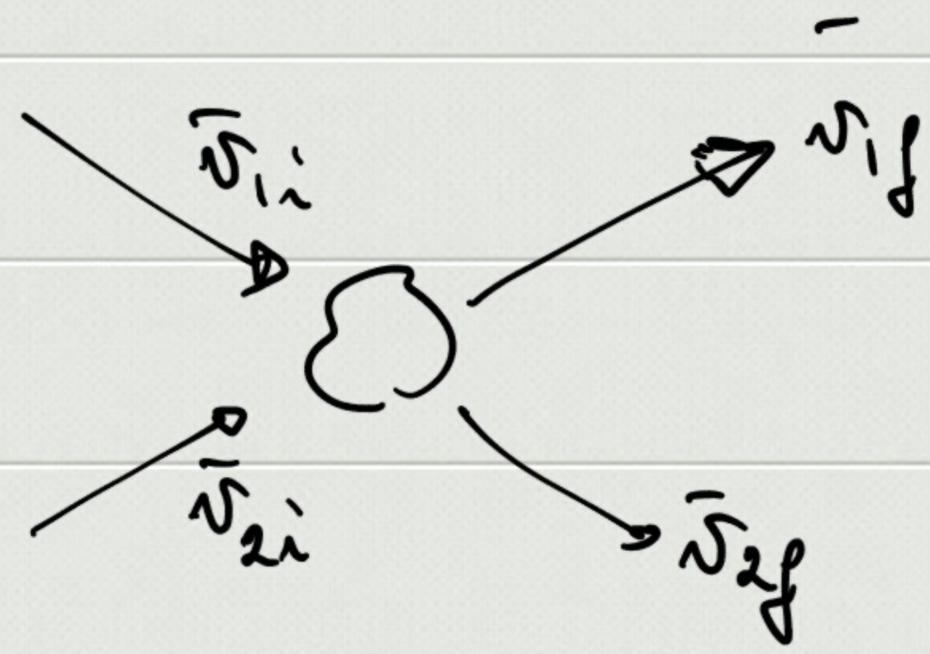
$$\begin{cases} \bar{P} = \text{cost} \\ E_k = \text{cost} \end{cases}$$



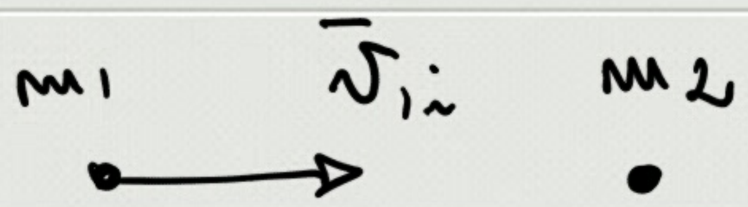
3D	4 equazioni	6 incognite
2D	3 "	4 "
1D	2 "	2 "

$$\begin{cases} m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \\ \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \end{cases}$$

$$\begin{cases} v_{1f} = \dfrac{(m_1 - m_2) v_{1i} + 2 m_2 v_{2i}}{m_1 + m_2} \\ v_{2f} = \dfrac{(m_2 - m_1) v_{2i} + 2 m_1 v_{1i}}{m_1 + m_2} \end{cases}$$

$$R = \frac{E_{k,2,f}}{E_{k,1,i}}$$

($\sqrt{R}$ coeff. di restituzione)



$v_{2i} = 0$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} \qquad v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

$$R = \frac{\cancel{\frac{1}{2}} m_2 v_{2f}^2}{\cancel{\frac{1}{2}} m_1 v_{1i}^2} = \frac{m_2}{\cancel{m_1}} \frac{\frac{4m_1^{\cancel{2}}}{(\quad)^2} \cancel{v_{1i}^2}}{\cancel{v_{1i}^2}} = \frac{4 m_1 m_2}{(m_1 + m_2)^2}$$

$\boxed{m_1 = m_2 = m}$ $\Rightarrow$ $v_{1f} = 0$ $\quad v_{2f} = v_{1i}$ $\quad R = 1$

$\boxed{m_1 >> m_2}$ $\Rightarrow$ $v_{1f} \cong \frac{m_1}{m_1} v_{1i} = v_{1i}$

$$v_{2f} \cong \frac{2\cancel{m_1}}{\cancel{m_1}} v_{1i} = 2 v_{1i}$$

$$R \cong \frac{4\cancel{m_1} m_2}{m_1^{\cancel{2}}} \simeq 0$$

$\boxed{m_1 << m_2}$ $\Rightarrow$ $v_{1f} \simeq \frac{-m_2}{m_2} v_{1i} = -v_{1i}$

$$v_{2f} \simeq \frac{2m_1}{m_2} v_{1i} \simeq 0$$

$$R = \frac{4 m_1 \cancel{m_2}}{m_2^{\cancel{2}}} \simeq 0$$